...on's correlation

$$P(x,y) = \frac{S_{xy}}{S_x \cdot S_y}$$

$$= \frac{\sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{n} (x_i - \bar{x})^2} * \sqrt{\sum_{i=1}^{n} (y_i - \bar{y})^2}}$$

$$\bar{x} = \text{mean}(x) = \frac{1}{n} \sum_{i=1}^{n} x_i$$

$$\bar{y} = \text{mean}(y) = \frac{1}{n} \sum_{i=1}^{n} y_i$$

n is the number of attributes in x and y

Ques:

DOC2 = ⟨3, 1, 4, 3, 1, 2, 0, 1⟩

DOC4 = ⟨0, 1, 0, 3, 0, 0, 2, 0⟩

Find pearson correlation index

Ans:

$x = \langle 3, 1, 4, 3, 1, 2, 0, 1 \rangle$

$y = \langle 0, 1, 0, 3, 0, 0, 2, 0 \rangle$

$$\therefore \bar{x} = \frac{3+1+4+3+1+2+0+1}{8} = \frac{15}{8} = 1.875$$

$$\therefore \bar{y} = \frac{0+1+0+3+0+0+2+0}{8} = 0.75$$

$$\sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y}) = (3-1.875) * (0-0.75) + (1-1.875) * (1-0.75) + (4-1.875) * (0-0.75) + (3-1.875) * (3-0.75) + (1-1.875) * (0-0.75) + (2-1.875) * (0-0.75) + (0-1.875) * (2-0.75) +$$

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$= (1-1.875) \times (0-0.75)$

$= ?$

$$\sqrt{\sum_{i=1}^{n}(x_i-\bar{x})^2} = \sqrt{(3-1.875)^2+(1-1.875)^2+(4-1.875)^2+(3-1.875)^2+(1-1.875)^2+(2-1.875)^2+(0-1.875)^2+(1-1.875)^2}$$

$= ?$

$$\sqrt{\sum_{i=1}^{n}(y_i-\bar{y})^2} = \sqrt{(0-0.75)^2+(1-0.75)^2+(0-0.75)^2+(3-0.75)^2+(0-0.75)^2+(0-0.75)^2+(2-0.75)^2+(0-0.75)^2}$$

$= ?$

$$\therefore p(x,y) = \frac{\sum_{i=1}^{n}(x_i-\bar{x})(y_i-\bar{y})}{\sqrt{\sum_{i=1}^{n}(x_i-\bar{x})^2}\sqrt{\sum_{i=1}^{n}(y_i-\bar{y})^2}}$$

$= ?$

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